

Method

Motivation of adaptive-weighted clip loss

The motivation of adaptive-weighted clip loss comes from a simple biological insights: Given a query cell, the differences between the query cell and different cell types should be different.

Suppose the query cell is $x = (x^{\{R\}}, x^{\{A\}}) \in \mathbb{R}^{r+a}$, and is belonged to cell type $C^{\{t\}}$. Then given cell x_1 also from $C^{\{t\}}$ and x_2 from different cell type $C^{\{q\}}$, then the relationship of differences between these two query pair should be

$$d(x, x_1) < d(x, x_2)$$

In original clip loss, we treated any embedding pairs from different cell pair as a negative pair, which ignores the internal biological structure. Given the query cell x and its different modality embedding $(z^{\{R\}}, z^{\{A\}})$, the clip loss maximize the difference between x and any other cell x_i while minimize the difference between its own two views $(z^{\{A\}}, z^{\{R\}})$, i.e.

$$\begin{aligned} \mathcal{L}^{a2r} &= -\log \frac{\text{sim}(z^{\{A\}}, z^{\{R\}})}{\sum_{i=1}^n \text{sim}(z^{\{A\}}, z_i^{\{R\}})} \quad \& \quad \mathcal{L}^{r2a} = \\ &= -\log \frac{\text{sim}(z^{\{R\}}, z^{\{A\}})}{\sum_{j=1}^n \text{sim}(z^{\{R\}}, z_j^{\{A\}})} \quad \& \quad \min \mathcal{L} = \\ &= \frac{\mathcal{L}^{r2a} + \mathcal{L}^{a2r}}{2} \end{aligned}$$

In order to take the biology structure into consideration, we proposed to distinguish the weights between different cell type pairs. Generally speaking, the differences should obey the following criteria:

$$d(x, y) < d(x, z) \quad ; \text{if } (x, y) \text{ belongs to the same cell type}$$

Therefore, we fomulate the clip loss into adaptive-weighted clip loss

$$\mathcal{L}^{a2r} = -\log \frac{\text{sim}(z^{\{A\}}, z^{\{R\}})}{\sum_{i=1}^n w_i \text{sim}(z^{\{A\}}, z_i^{\{R\}})}$$

In practice, it is hard to estimate the cell-cell difference weights $w_{\{ij\}} \in \mathbb{R}^{+}$, we introduce a adaptive weight strategy. The details of the algorithms are listed as following:

1. compute the original similarity matrix $C = (C_{\{ij\}}) \in \mathbb{R}^{n \times n}$, in which $C_{\{ij\}} = z^{\{A\}}_i \cdot z^{\{R\}}_j$
2. for each row, we split the column indices into 3 part. $\mathcal{N}_l \cup \mathcal{N}_h \cup \mathcal{N}_m$, which corresponds to low(l , the top- k weight), high(w , the last top- k weight), medium(m , other weight), confidence negative samples.

3. compute the row weighted clip loss

$$\mathcal{L}^{a2r} = -\log \frac{C_{\{ij\}} \{w_l \sum_{i \in \mathcal{N}_l} C_{\{ij\}} + w_h \sum_{i \in \mathcal{N}_h} \text{sim}(z^{\{A\}}, z_i^{\{R\}}) + w_l \sum_{i \in \mathcal{N}_l} \text{sim}(z^{\{A\}}, z_i^{\{R\}})\}}{}$$

4. for each columns, perform the similar split and weighted sum.
5. average the row weighted clip loss and column weighted clip loss.

The theoretical analysis of the adaptive-weighted clip loss

We claim that the adaptive-weighted clip loss not only considering the biological structure, but also can xxxx consider a single negative pair (i, j) , we omit the upper font for brevity. The derivatives corresponding to (i, j) is

$$\begin{aligned} & \frac{\partial \mathcal{L}}{\partial c_{ij}} = \frac{\exp(c_{ij})T_{-j} + \exp(c_{ij})}{\sum_{l \neq j} \exp(c_{il})} \& T_{-j} = \end{aligned}$$

For the adaptive-weighted clip loss, suppose we only modify the weight of (i, j) , then we have

$$\begin{aligned} & \frac{\partial \mathcal{L}^w}{\partial c_{ij}} = \frac{w_{ij} \exp(c_{ij})}{w_{ij} \exp(c_{ij})} T_{-j} + \end{aligned}$$

Then we have

$$\begin{aligned} & \frac{w_{ij} \exp(c_{ij})}{w_{ij} \exp(c_{ij})} T_{-j} + w_{ij} \exp(c_{ij}) > \frac{\exp(c_{ij})}{\sum_{l \neq j} \exp(c_{il})} T_{-j} + \exp(c_{ij}) ; ; \\ & \text{if } w_{ij} > 1 \& \frac{w_{ij} \exp(c_{ij})}{w_{ij} \exp(c_{ij})} T_{-j} + w_{ij} \exp(c_{ij}) < \frac{\exp(c_{ij})}{\sum_{l \neq j} \exp(c_{il})} T_{-j} + \exp(c_{ij}) ; ; \\ & \text{if } w_{ij} < 1 \end{aligned}$$

Therefore, for (i, j) from the similar pair, we set $w_{ij} < 1$, then the gradient update following

$$c_{ij}^{\text{new}} = c_{ij} - \frac{\partial \mathcal{L}^w}{\partial c_{ij}} > c_{ij} - \frac{\partial \mathcal{L}}{\partial c_{ij}} = c_{ij}^{\text{new}}$$

i.e., for low confidence negative sample, their similarity c_{ij} decrease slower than original clip update.

For (i, j) from the different pair, we set $w_{ij} > 1$, then the gradient update following

$$c_{ij}^{\text{new}} = c_{ij} - \frac{\partial \mathcal{L}^w}{\partial c_{ij}} < c_{ij} - \frac{\partial \mathcal{L}}{\partial c_{ij}} = c_{ij}^{\text{new}}$$

i.e., for high confidence negative sample, their similarity c_{ij} decrease faster than original clip update.

Overview of Solid Recover model

The Solid Recover model

Single omic Encoder

We construct a single omic encoder using MLP. A single block is formulated as following:

Alignment module

Single omic Decoder

Training Loss

Evaluation of multi-modality integration

foscttm top-k hit rate

Evaluation of embedding quality

ARI NMI

Evaluation of cross modality generation

Evaluation of in-sicilo perturbation

Construction of Mouse Brain Vector Database